Type conversion and number system

PATEL WEB SOLUTION

We believe in quality

Agenda

1. Type conversion
2. Number System
3. Conversion of number system
4. unicode

Type Conversion

number = int, float, complex, bool

a=5 → int
b="5" → str

a+b   int + str   Error

a+int(b)   int + int   No error

10 → int

str(a)+b   str + str   No error

"55" → str

Type conversion functions

int( )
float( )
complex( )
bool( )
str( )

Number System

Binary Number System – 0, 1

Octal Number System – 0, 1, 2, 3, 4, 5, 6, 7

Decimal Number System – 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Hexadecimal Number System – 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

Counting

Dec	Oct	Hex	Bin
0	0	0	0
1	1	1	1
2	2	2	10
3	3	3	11
4	4	4	100
5	5	5	101
6	6	6	110
7	7	7	111
8	10	8	1000
9	11	9	1001
10	12	A	1010
11	13	B	1011
12	14	C	1100
13	15	D	1101
14	16	E	1110
15	17	F	1111
16	20	10	10000
17	21	11	10001
18	22	12	10010
19	23	13	10011
20	24	14	10100

Dec	Oct	Hex	Bin
21	25	15	10101
22	26	16	10110
23	27	17	10111
24	30	18	11000
25	31	19	11001
26	32	1A	11010
27	33	1B	11011
28	34	1C	11100
29	35	1D	11101
30	36	1E	11110
		1F	
		20	

place value

Decimal

$$2148 = 2000 + 100 + 40 + 8$$

$10^3 \quad 10^2 \quad 10^1 \quad 10^0$

Octal

$$3145 = 3 \times 512 + 1 \times 64 + 4 \times 8 + 5$$
$$= 1536 + 64 + 32 + 5$$
$$= 1637$$

$8^3 \quad 8^2 \quad 8^1 \quad 8^0$

Hexadecimal

$$A10F = A \times 16^3 + 1 \times 16^2 + 0 \times 16^1 + F$$

$16^3 \quad 16^2 \quad 16^1 \quad 16^0$

Binary

$$1001110 = 2^6 + 2^3 + 2^2 + 2^1$$

$2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

Conversion of Number System

Dec → Bin

$25 \rightarrow 11001$

$$
\begin{array}{r|l|l}
2 & 25 & \\
\hline
2 & 12 & 1 \\
\hline
2 & 6 & 0 \\
\hline
2 & 3 & 0 \\
\hline
2 & 1 & 1 \\
\hline
 & 0 & 1
\end{array}
\quad \uparrow
$$

	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
	128	64	32	16	8	4	2	1
25				1	1	0	0	1
76		1	0	0	1	1	0	0
100		1	1	0	0	1	0	0

1 1 0 0 0 1 0 0 = 196

128 64 4

1 0 1 0 1 0 1 = 85

64 16 4 1

Number System Conversion functions

$x = 25$

bin(x) → '0b11001'

oct(x) → '0o31'

hex(x) → '0x19'

$x = 0b11001$

$x = 0o31$

$x = 0x19$

Unicode

The Unicode is character encoding, and its goal is to replace the existing character sets with its standard UTF.

UTF - Unicode Transformation Format

UTF-8 is the most commonly used character encoding.
It is also backward compatible with ASCII

character to unicode

x = 'A'

ord(x) → 65

unicode to character

x = 65

chr(x) → 'A'